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Research article

The loss of political connections and the fluctuation of corporate stock prices: Moderating effect based on media attention

Qiong Sun [a], Qingzhou Xiao [a], Jingjing Jiang [a,*], Xiankai Huang [b]

[a] Beijing Union University, Beijing 100101, China
[b] Beijing Technology and Business University, Beijing 100048 China

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Abstract:
The loss of political connections
Stock price fluctuations
Media attention
The resignation of "official" independent directors

ABSTRACT

This article examines the impact and mechanism of political connections on stock price fluctuations after the resignation of independent directors with "official" status, based on the exogenous influence of Document No. 18 of the Central Organization Department. Using panel data of A-share listed companies on the Shanghai and Shenzhen stock markets from 2012 to 2020, the experimental group and control group for political affiliation deficiency events were determined through propensity score matching (PSM), and a quasi natural experiment was constructed using differences in differences (DID) for empirical research. By examining the relationship between regional financial development, lack of political connections, and stock price fluctuations, we found that regions with higher levels of financial development are more prone to stock price fluctuations due to the lack of political connections, which is related to higher levels of debt financing in regions with higher levels of financial development. In addition, the increasing level of debt exacerbates conflicts and inconsistencies among stakeholders, which is not conducive to the stability of the company's stock price. Through the above research, relevant suggestions have been provided to enterprises, media, and governments. For example, enterprises should clarify the boundary between government and enterprise, focus on long-term strategic goals, build core competitiveness, and thereby enhance their own value; The media should play a correct role in information transmission and public opinion guidance, and play a positive role in the economic development of the industry; The government should promote market-oriented construction and establish positive government enterprise relationships.

1. Introduction

With the continuous development of the Chinese economy, the dependence of enterprises on political connections is also deepening. In this process, political connections not only have a profound impact on the development of enterprises, but also pose new challenges to their governance structure. On the one hand, political connections can help enterprises obtain more policy support and resource guarantees, and improve their competitiveness; On the other hand, excessive political connections may lead to complexity in the decision-making process of enterprises, and even affect their market image and reputation [1]. In China, the government has a high degree of control over resource allocation, which forces companies to seek close ties with the government to obtain more resources [2]. However, this political connection often accompanies companies catering to the will of the government, which may lead to companies

* Corresponding author.
E-mail address: jingjingjiangbuu@163.com (J. Jiang).

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deviating from market orientation and reducing their operational efficiency [3]. In addition, political connections may also lead companies into political turmoil, posing potential risks to them. In terms of information disclosure, political connections make it difficult for enterprises to fully comply with market rules and laws and regulations in the process of interacting with the government. This may not only lead to opacity in corporate information disclosure, but also result in companies taking unfair measures to maintain their political connections when facing regulation [4]. This phenomenon not only damages the fair competition environment in the market, but also provides a breeding ground for corruption. The impact of political connections on corporate stock prices is also significant [5]. In the context of political connections, enterprises often receive special policy support and resource protection, which makes investors have high expectations for such enterprises, thereby driving up the stock price of the enterprise [6]. However, this rise in stock prices may not be based on the actual business performance of the company, but rather on expectations of political connections. Once a company is unable to fulfill this expectation, the stock price may experience significant fluctuations, causing losses to investors [7].

The news media, as an effective tool for disseminating information, collects and disseminates data on the market economy and enterprises to the public, and influences the behavior of enterprises and investors through the broadcast of information. In a limited capacity, the news media also serve as watchdogs for corporations, so protecting investors [8]. Other experts' research has demonstrated that politically connected companies are more likely to obtain media coverage. Thus, media coverage of capital market involvement is a significant element influencing investors' decisions and the share prices of politically related firms [9].

Since the 18th National Congress of the Communist Party of China, the state has implemented a range of strategies to optimize the market environment regularly. Among these, the 2013 ″Document No.18″ released by the Organization Department of the Central Committee of the CPC ("Opinions on Further Regulating Party and Government Leading Cadres' Part-time Jobs in Enterprises") has had a significant impact. The paper states that party and government officials beyond a particular level, whether they are active officials or have retired within the past three years, are not permitted to hold part-time or full-time roles in any firm. In addition, officials who have been retired for at least three years are permitted to occupy roles in businesses [10]. The release of the document precipitated a wave of widespread resignations among government directors. Companies with political ties to government officials' directors consequently suffered a loss of those ties. Since nearly a decade has passed since the policy's inception, we are now in a position to observe the effect of the loss of political connections on stock prices. In addition, previous to this study, the majority of studies examining the association between the loss of political connections and stock prices utilized the event study methodology and reached divergent conclusions.

In order to understand the impact of government policies on the Chinese capital market, this article establishes a quasi natural experiment based on the lack of political connections between enterprises and the withdrawal of "official" independent directors. Considering sample selection bias and endogeneity, this study used propensity score matching (PSM) to empirically study the impact of political connectivity loss on stock price volatility, and further tested the moderating effect of media attention. Analyze the impact of lack of political connections on stock price fluctuations, providing reference for creating a favorable stock market environment and political business relationships. Secondly, this article also studied the impact of political connection losses under media influence on stock price fluctuations. The structure of this study is as follows: The first section analyzes the relationship between the lack of political connections in enterprises, stock volatility, and media attention through literature review and theoretical discussion, and proposes research hypotheses; The second section introduces the research design and data sources; The third section reports the analysis of empirical results, mainly including propensity score matching results and descriptive statistics, benchmark regression analysis, moderation effect analysis, robustness testing, and further analysis; The fourth section presents relevant conclusions and suggestions.

2. Literature review and hypothesis Formulation

2.1. Literature review

2.1.1. The loss of political connections and the fluctuation of corporate stock prices

In several countries across the global, political connections and their study are frequent [11]. Domestic and foreign experts have performed study on the resource advantages that political affiliation confers on several facets of business development. In terms of innovation strategy, some academics feel that political connections boost corporate innovation due to benefits such as access to financial information [12], and other academics feel that political connections may also affect resource allocation and long-term investment strategies, so diverting firm investment activities from corporate innovation [13]. In terms of investment and funding, some academics believe politically connected businesses tend to have higher investment levels [14]. At the same time, it can help alleviate the finance limitations of enterprises [15]. Regarding manufacturing and operation, access to the capital market and critical government resources, favorable regulatory treatment and government contracts [16], will also lower worker costs, whereas the disintegration of political connections will result in increasing employee compensation and costs. Nonetheless, regardless of whether it is innovation strategy, manufacturing and operation, or investment and finance behavior, each has a significant impact on the business value and stock price. Because of this, academics are slowly beginning to value the connection between political connections and firm stock values.

The influence of political connections on the stock prices of corporations is multidimensional. Some scholars believe that political connections can be used as a property rights protection mechanism so that companies can avoid the risk of stock price collapse [17], and some scholars believe that companies actively disclose bad news about company operations in the process of seeking political connections [18], which gives stakeholders confidence and reduces the risk of corporate stock price collapse through active disclosure of negative information. From the perspective of resource dominance and competition, some academics have investigated the effect of political connections on the stock price of enterprises. The study indicated that political connections in state-owned enterprises have a greater impact on the stock price than in private enterprises [19]. With the adoption of a succession of measures that assist the optimization of the market environment, domestic scholars have shifted their attention to the influence of the loss of political connections on the share prices of corporations. About fifty percent of companies in which official independent directors serve have not created positive value, according to Shao et al. [20] on the impact of the resignation of official independent directors. Studying the effect of loss of political connections on corporate stock prices, marked by the resignation of official independent directors, reveals that loss of political, based on the perspective that outside investors care about information transparency, reduces the risk of corporate stock price collapse due to weakened rent-seeking incentives and increased information transparency [21]. The loss of political connections, however, has a detrimental impact on corporate stock prices from the viewpoint of how outside investors view and perceive access to corporate resources [22].

2.1.2. The loss of political connections, media attention, and company stock prices

Through the mining and dissemination of information, the media has an effect on the company's share price. From the perspective of the purpose of media coverage, in addition to the media's own characteristics, Liu and Mo [23] believed that companies may plan to "hype" in order to attract investors. This is in reference to the company's stock price and the relationship between its corporate media coverage. In terms of the influence, Tan et al. [24] have investigated and confirmed that the media, when providing information to the market, not only provides updated information quickly and in a timely manner, but also may bring the media sentiment contained in the information to investors, thus attracting investors' attention or resulting in behavioral biases. According to some scholars, politically connected businesses are more likely to be impacted by media attention. Political connections also have a greater impact on corporate stock issuance, financing, and other behaviors in the context of media attention [25]. Wang et al. [26] have demonstrated that corporations with political connections exhibit controlling behavior that is less susceptible to the media's effect than do enterprises without political connections. In addition, the role of media attention is constrained by the "administrative influence" that political connections have over enterprises, and on the other hand, the beneficial effects of media attention on information disclosure are weaker in enterprises with strong political connections than in enterprises without political connections [27].

2.1.3. Review

As a result, no consensus has been reached regarding how political resources affect the company's information environment and stock price stability [21]. Political connections, stock price volatility, and media attention are all closely related. When we look at past research done by scholars, we can see that these studies either study the relationship between political connections and stock prices or the relationship between media attention and stock prices [24]. Few scholars study the relationship between "political connection - media attention - stock price volatility" within a single theoretical framework, particularly when looking at the relationship between the erosion of political connections caused by media coverage and company stock price volatility. There are still open questions regarding how the loss of political connections influences stock price volatility and how the media acts as an external information source after the loss of political connections, and this study offers some empirical evidence for these questions.

In summary, this study conducted in-depth analysis from three aspects: political relations, media attention, and company stock price fluctuations, revealing their interrelationships. The research results have certain enlightening effects on enterprises and governments, helping to improve their understanding and grasp of stock price fluctuations, and thus providing useful references for decision-making of enterprises and governments. However, this study still has certain limitations, and future research can further expand and deepen related fields, in order to provide more comprehensive theoretical support for the governance of company stock price fluctuations.

2.2. Formulation of hypothesis

2.2.1. The loss of political connections and stock price fluctuation

Official independent directors have a significant impact on enterprises in the following ways as a vital corporate political connection [28]. First, according to the resource-based theory, enterprises that forge political connections have an advantage in accessing outside resources, such as knowledge, money, technology, and other resources. Numerous domestic and international research have demonstrated that businesses with political connections have a higher likelihood of obtaining bank loans [29]. Enterprises that have developed political connections are more likely to collaborate with academic institutions of higher learning in order to get the technical resources necessary to foster innovative business practices and enable enterprises to experience rapid development. Because of the interaction between independent directors and the government, businesses are able to access more information resources, including public decision-making, economic development priorities, and direction. Second, the political connections the company has made give it a good institutional setting in which to request from the government a favorable external development environment. This includes requesting administrative authority for the company to enter a barrier industry and broaden its business. Political connections can also help businesses in highly monopolized industries obtain priority and keep a sizable market share in the absence of core competencies. A positive connection with the government makes it simple to encourage enterprises to apply for tax incentives [30]. Additionally, because of their connections to the government, businesses frequently exhibit proactive social responsibility, enhancing their reputation. Additionally, because they have strong external relationships, independent directors with significant ties to the government can aid businesses in resolving crises [31]. The government may provide direct subsidies, mergers and acquisitions, and other forms of assistance to a firm when it confronts significant dangers. Third, from the standpoint of internal corporate governance, executives with corporate political connections stifle the dissemination of bad news, such as the exposure of unlawful information, in order to support the government's image of economic boom. As a result, under the benefits of resources and an institutional environment given by political connections for enterprises, the value, operation, and financial indicators of enterprises have demonstrated a consistent development trend in a positive direction, and the signals issued are conducive to increasing investors' confidence in holding shares.

However, the loss of political connections and the resignation of independent directors with "official" status make it more difficult for the company to secure outside development and may even weaken its core competitiveness. This has a negative effect on the company's development and communicates information that is not helpful for the long-term success of the business. Additionally, research has indicated that firms with political relations have a greater degree of effect on their information disclosure during the development of their operations than firms without political affiliations [32]. Losing political connections prevents businesses from concealing information, which has a bigger influence on stock prices. Additionally, it will lessen the likelihood that the business will receive backing from stakeholders like shareholders, which will undermine the stability of the stock price. Therefore, we make the following assumptions.

H1. The loss of political connections has a significant positive impact on stock price fluctuation.

2.2.2. The moderating effect of media attention

Many scholars have found evidence to support the view that media attention may affect the financing methods of enterprises. However, media reports also have a certain hype effect, causing investors to turn a blind eye to their investment behavior. Overall, the role of the media in disseminating information is greater than that of a company's formal information disclosure channels, and its impact on disseminating information is broader and faster. The information in the media is more likely to affect the emotions of investors, thereby influencing their investment decisions, especially when the media is emotional. In addition, evidence from the Chinese stock market suggests that media coverage often leads to stock prices deviating from their fundamental value levels. On the one hand, when commercial transparency is limited, media attention has a greater impact on the degree of deviation in stock value, as the media is a key participant in the information transmission process [33]. On the other hand, in a company with high transparency, media attention increases the transparency of company related information, which is quickly conveyed to investors and other stakeholders under media attention, affecting their behavior in investment or cooperation. The original political connections of the company were lost due to the widespread resignation of independent directors, and this signal was conveyed to investors under the influence of the media. In addition, due to the self-awareness or cognitive bias of the media, the "information effect" of such media may generate a certain degree of information divergence. This deviation may be the result of media hype caused by cooperation between competitors. In short, when the media pays attention to a company, such as during news consulting, the impact of losing political ties on the company can be exaggerated or even distorted. After obtaining information through the media, investors are more likely to be influenced by the content of the information, which has an impact on stock prices. Therefore, this study proposes the following hypotheses.

H2. Media attention has a positive moderating effect on the relationship between political connection loss and stock price volatility.

3. Research design

3.1. Samples and data

With the exception of financial and insurance firms, ST companies, and some samples with missing values, this study uses panel data from A-share listed companies in Shanghai and Shenzhen from 2012 to 2020 as its research sample. As a test for the removal of political ties, information from firm announcements regarding the resignation of "official" independent directors was manually gathered using the Wind database. The time of the mass resignation of "official" independent directors was ultimately determined to be 2014, and a total of 1022 samples of enterprises with independent director resignations were obtained. The document was promulgated at the end of 2018, taking the implementation time of the policy into consideration. A total of 409 of them made it plain in their resignation announcements that they did so because of "Document No. 18″ and its associated restrictions. This study further identifies each of these independent directors in light of the ambiguity in the disclosure of their resignation for other reasons, such as personal or professional ones. This study will distinguish the independent directors who work for the government, public institutions, and state-owned enterprises due to the "Document No. 18″ in order to more precisely define the political identities of independent directors in political connections. Since social networks of former public servants continue to exist even after they leave government, they act as a link between business and government and government companies in gaining various financial benefits from the latter. At the same time, drawing on the practice of Deng et al. [22], "official" independent directors are defined as those who are currently or have served in central or local government departments, courts, and procuratorates at all levels, or serve as the director and deputy director of the National People's Congress, as well as chairmen or vice chairmen of the Chinese People's Political Consultative Conference. Additionally, this study makes reference to Yuan et al. [34] research and only keeps the sample of independent directors who leave their positions as officials at the bureau level and above in order to ensure the effectiveness of political connections. As a result, 260 enterprises are included in the study's sample. The CSMAR database, Sina Finance, and Taoguba are the key sources of information on independent directors' identities. The China Financial Yearbook provides the information on the level of regional financial development. In addition to other data, the CSMAR database also contains information about stock price fluctuations. The continuous variables, with the exception of the dummy variables, were processed with winsorize at the 1% level to assure the accuracy of the experimental results.

3.2. Models and variables

This study employed the propensity score matching method (PSM) to identify the experimental group in which political association loss happened and the control group in which political association loss did not occur in order to precisely match the experimental and control groups and to avoid endogenous interference. To measure the propensity scores of the experimental and control group variables, Logit regression was specifically used. Additionally, the Mahalanobis distance matching approach is employed for sample matching in accordance with the propensity score. The study also used differences-in-differences (DID) to experimentally investigate the effect of the loss of political connections on stock price volatility brought on by the resignation of independent directors with official status brought on by the "Document No. 18".

The basic econometric model setup of this paper [34] is constructed as in Formula (1).

$$VOL_{it} = \beta_0 + \beta_1 treated_{it} \bullet time_{it} + \beta_2 treated_{it} + \beta_3 time_{it} + \beta_4 control_{it} + \beta_5 Idcode_{it} + \beta_6\ Year_{it} + \varepsilon_{it} \tag{1}$$

treaded is the firm grouping dummy variable. It is 1 if the firm experienced a loss of political affiliation and 0 otherwise. *time* is a dummy variable for the time when a firm experiences a loss of political affiliation. After the loss of political connection occurs, i.e. in 2014–2020; before the loss of political connection, i.e. in 2012–2013. The coefficient β_1 of the product of the two is the treatment effect of the interaction term.

In the model used to evaluate stock price volatility, the key explanatory variable is called VOL, which is annualized volatility. In the calculation, it is chosen to obtain the logarithm by adding 1 to represent the annualized volatility. Based on the research of Cheng and Zhang [35], the specific calculation process includes dividing the sample data into multiple sample intervals on a monthly basis, and then calculating the standard deviation of the average return rate for each interval. Using monthly stock data to calculate the standard deviation Mstdev of monthly returns, the annualized volatility can be expressed in Formula (2):

$$VOL = \sqrt{\text{Mstdev} \times 12} \tag{2}$$

$control_{it}$ represents a control variable. According to Dong and Wu, equity concentration as a crucial form of corporate internal governance has a substantial effect on firm share prices [36]. Xiao and Xia [37] demonstrated that the characteristics of the board of directors, such as the shareholding ratio of the board of directors and the combination of two positions, will affect the freedom of innovation of enterprises, which will then affect the stability of enterprises in the constantly shifting market environment. In the process of corporate operation and development, company size, listing time, return on assets, asset-liability ratio, intangible assets, etc., are also significant elements that influence stock price fluctuations [38]. Additionally, He et al. [39] thought that stock price anomalies could be caused by the turnover rate. Therefore, this study selects company size, company age, and return on assets, asset-liability ratio, intangible assets, board shareholding ratio, ownership concentration and turnover rate as control variables. $Idcode_{it}$

is the individual fixed effect, $Year_{it}$ is the time fixed effect, and ε_{it} is the random disturbance term. The variable abbreviations and variable meanings of the control variables are shown in Table 1.

The model setting structure of the moderation effect test of media attention [39] is shown in Formula (3):

$$VOL_{it} = \beta_0 + \beta_1 treated_{it} \bullet time_{it} + \beta_2 treated_{it} + \beta_3 time_{it} + \beta_4 control_{it} + \beta_5 treated_{it} \bullet time_{it} \bullet MF_{it} + \beta_6 MF_{it} + \beta_7 Idcode_{it} + \beta_8 Year_{it} + \varepsilon_{it} \tag{3}$$

In Formula (3), MF represents the media attention, referring to Chen [40], using two methods of newspaper attention and network media attention to represent the media attention. The first is media coverage in newspapers. In this study, the number of newspaper financial news stories reporting news of listed businesses is collected using the CNKI as the data source. The second is the focus of online media. To count the number of news stories mentioning listed firms in the news headlines of listed companies, this study uses the Baidu news search engine. The number of news stories with listed firms in the headlines of financial news articles in newspapers and news stories with listed companies in the headlines of online media is the total of the media attention. This study logarithmically processes the number of statistics news reports and uses the natural logarithm of the number of media reports plus 1 as a proxy for media attention in order to minimize orders-of-magnitude interference. The moderating impact of media attention on the loss of political association and stock price volatility is represented by the interaction term's coefficient.

Table 1
Definition of control variables.

Symbol	Name of the variable	Meaning of the variable
age	Age of the company	The difference between the current year and the year the company was listed
Top1	Shareholding concentration	Shareholding ratio of the largest shareholder
boardshareholdingratio	Board of directors shareholding ratio	Shareholding ratio of the company held by the board of directors
ROA	Profitability	Operating Profit/Total Assets
leverage	Financial leverage	Total Liabilities/Total Assets
lnintangibleassets	Intangible assets	The natural logarithm of the total intangible assets of the enterprise
lnsize	Enterprise size	The natural logarithm of the firm's total assets
ToverTlY	Annual turnover rate	Sum of daily turnover (total shares) during the year

4. Analysis of empirical results

4.1. *Empirical results and analysis using 2014–2020 as the experimental period*

4.1.1. *Propensity score matching results*

We finally obtained 19,469 observations within the shared support domain by using propensity score matching, including 2266 in the experimental group and 17,203 in the control group. The kernel density function map is used to evaluate initially, increasing the reliability of the propensity score matching results. The matching effect is improved by the amount of kernel density map overlap between the experimental group and the control group. The kernel density in Fig. 1 is depicted before and after propensity score matching (PSM). The kernel density function curves of the control group and the experimental group practically coincide from the starting point to the end after propensity score matching (PSM), which shows that the matching is successful, as can be seen in Fig. 1.

In order to determine whether the results of propensity score matching (PSM) meet the requirement of conditional independence, additional balance tests were conducted on the PSM results, as shown in Table 2. Firstly, the matching effect is good, manifested as the absolute standard deviation of the matching variables being less than 20 and close to 0. Secondly, after propensity score matching (PSM), the T-value of the T-test is no longer significant, indicating that the null hypothesis that the means of the matching variables are equal is accepted, thus verifying the validity of the propensity score matching (PSM) results.

4.1.2. *Descriptive statistics*

Table 3 presents the descriptive statistical results of the main variables for a complete sample of 19230 observations after further removing missing values and propensity score matching (PSM). Table 3 shows that there are significant differences in stock price fluctuations between different companies and years, which may be related to factors such as the industry and development level of the companies. The selected sample covers enterprises from different periods, and there is a time lag between the initial data disclosure and the annual report disclosure of specific enterprises, in order to select matching samples as much as possible. The shareholding ratio of the board of directors varies greatly among different enterprises, which may be due to the nature of their business. According to relevant financial indicators, all enterprises have shown normal operation, indicating that since the optimization of the macro market environment in 2012, the overall development momentum of listed companies in China has been strong.

4.1.3. *Benchmark regression results*

The major effects of the model with double differences were estimated in this investigation using Stata17.0, and the findings are displayed in Table 4. It is obvious from the benchmark regression findings that the interaction term $treaded \times time$ is not significant. This might be because the volatility of the stock price of the company is only temporarily impacted by the loss of political connections brought on by the resignation of independent directors. Moreover, as a result of the continual optimization of my country's market environment and the implementation of a series of government policies that are advantageous to the survival and development of enterprises, the economy is operating smoothly, and the stock values of enterprises are generally steady under the influence of the macro environment as a whole.

4.2. *Empirical results and analysis of the shortened experimental period (2014–2016)*

Shorten the event impact interval and shorten the time in order to further confirm the mechanism of the loss of political link on stock price variations. This study shortens the experimental period to 2016 and performs propensity score matching (PSM) once more while creating a quasi-natural test (DID) to examine the factors affecting stock price volatility.

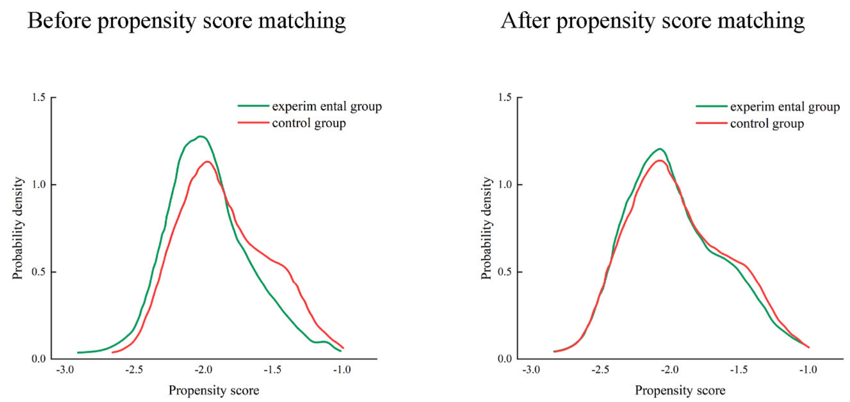

Fig. 1. Kernel density function diagram
Before propensity score matching
After propensity score matching.

Table 2
Balance test results.

Matching variable	Before matching	Mean value		Standard deviation (%)		T-test
	After matching	Experimental group	Control group	Deviation	Reduction	P>ItI
age	U	13.072	12.184	13.0		0.000
	M	13.072	13.287	−3.1	75.8	0.292
Top1	U	36.705	33.732	19.6		0.000
	M	36.705	36.705	0.0	100.0	0.999
boardshareholdingratio	U	0.07931	0.10013	−12.8		0.000
	M	0.07931	0.08007	−0.5	96.4	0.870
ROA	U	0.0367	0.03363	4.8		0.035
	M	0.0367	0.03688	−0.3	94.4	0.923
leverage	U	0.46627	0.44327	11.0		0.000
	M	0.46627	0.46586	0.2	98.2	0.947
lnintangibleassets	U	19.274	18.717	28.7		0.000
	M	19.274	18.74	−2.3	92.0	0.421
lnsize	U	22.775	19.253	29.7		0.000
	M	22.775	22.763	1.2	95.9	0.686
ToverTlY	U	3.5097	3.8894	−13.8		0.000
	M	3.5.097	3.4953	0.5	96.2	0.854

Table 3
Descriptive statistics results.

Variable	Observed value	Mean value	Median	Standard deviation	Minimum value	Minimum value
VOL	19230	1.206	0.274	0.827	0.705	2.130
FD	19230	3.688	1.520	2.091	1.828	8.131
MF	19230	144.908	149.886	5.000	0.000	893.000
boardshareholdingratio	19230	0.098	0.167	0.000	0.000	0.632
age	19230	12.254	6.992	2.000	1.000	26.000
ROA	19230	0.034	0.065	−0.067	−0.283	0.212
top1	19230	34.051	14.923	13.177	8.482	74.824
leverage	19230	0.445	0.210	0.115	0.054	0.945
lnintangibleassets	19230	18.813	1.784	15.831	12.696	23.452
lnsize	19230	22.409	1.306	20.527	19.726	26.364
ToverTlY	19230	3.848	2.797	0.800	0.370	14.090

Table 4
Benchmark regression results.

Variable	Stock price fluctuation
treated×time	−0.0085382
	(-0.81)
age	0.039414*
	(3.97)
Top1	0.0012168*
	(4.68)
boardshareholdingratio	0.1277364*
	(5.43)
ROA	0.1129893*
	(3.73)
leverage	0.204105
	(1.27)
lnintangibleassets	0.000316
	(014)
lnsize	−0.104572**
	(-2.30)
Idcode	Control
Year	Control
_cons	0.9095619***
	(7.61)

Note: *,**,**denote 10%, 5%, and 1% significance levels, respectively. The t-statistic is shown in parentheses. The same below.

4.2.1. *Propensity score matching results*

We finally collected 10,792 observations within the same support domain using propensity score matching, including 1259 in the experimental group and 9533 in the control group. The kernel density function curves of the control and experimental groups almost overlapped from the starting point to the end point after propensity score matching (PSM), as shown in Fig. 2, indicating that the effect of this propensity score matching was still good after examination by kernel density function plots.

In order to verify whether the propensity score matching (PSM) results are consistent with the conditional independence hypothesis, the PSM results were further evaluated for equilibrium; the results of these tests are provided in Table 5. The matched variables' absolute standard deviation was still around 20 and very near to 0. The propensity score matching (PSM) results were still valid, as shown by the T-value of the T-test conducted after the PSM, which was likewise no longer significant.

4.2.2. *Benchmark regression results*

The outcomes of the model's benchmark regression are reported in Table 6 after the major effects of the model were double-difference estimated. The coefficient of the interaction term is substantially positive at the 0.1% level ($\beta_1 = 0.0033194, p < 0.01$), according the findings of this benchmark regression. The study's findings indicate that political resources have a stabilizing influence on the stock prices of Chinese listed firms once the impact period following the incident is reduced. The initial competitive advantage has diminished or completely vanished as a result of the loss of political connections, which has an impact on investment decision-making, alters investor behavior, and raises stock price volatility. This supports the H1 hypothesis.

4.2.3. *Moderating effect analysis*

In order to test the moderating effect of media attention on the relationship between the loss of political connections and the volatility of corporate stock prices, this study introduces the interaction term $treated \times time \times MF$ between the loss of political connections and media attention, and regresses model (3). The regression results are shown in Table 7. The results show that the three interaction coefficients are significantly negative at the 5% level ($\beta_5 = -0.0000102$, $p < 0.05$), indicating that companies with higher media attention have less impact on stock price volatility after the loss of political connections, and reject the H2 hypothesis. This might be as a result of the media's propensity to report on favorable news stories regarding the introduction of policies and the details of their implementation. Particularly powerful official media, including the press and official financial news, contribute to the improvement of the market environment. By demonstrating to investors that businesses have improved their core capabilities and are better integrated into the market environment following a loss of political affiliation, they boost investor confidence. Higher media attention, on the other hand, acts as a watchdog and an incentive for management with political affiliation, encouraging them to change their plans as soon as possible and put their attention on the long-term sustainable development of the business, effectively reducing the negative impact on the stock price and reducing the risk of a stock price crash.

4.2.4. *Robustness tests*

Prior usage of propensity score matching (PSM) established the necessary conditions for the subsequent double difference to satisfy the parallel trend hypothesis test. Consequently, the parallel trend test does not require repetition in the robustness examination. This study employed a placebo test to generate randomized trials and regressed them in accordance with Table 6 to assess the extent to which the aforementioned outcomes are influenced by random factors, omitted variables, etc. The validity of the results was assessed based on the likelihood of obtaining the baseline regression's estimated coefficients from the fictitious experiment. The distribution of the fictitious regression coefficients was determined after 500 iterations of the aforementioned process, as illustrated in Fig. 3. The estimated coefficients of the fictitious double difference term are concentrated around 0, as depicted in the figure, indicating that there is no significant omitted variable issue in the model setup and that it is unaffected by other random factors. Consequently, the core conclusion robustness test is passed once more.

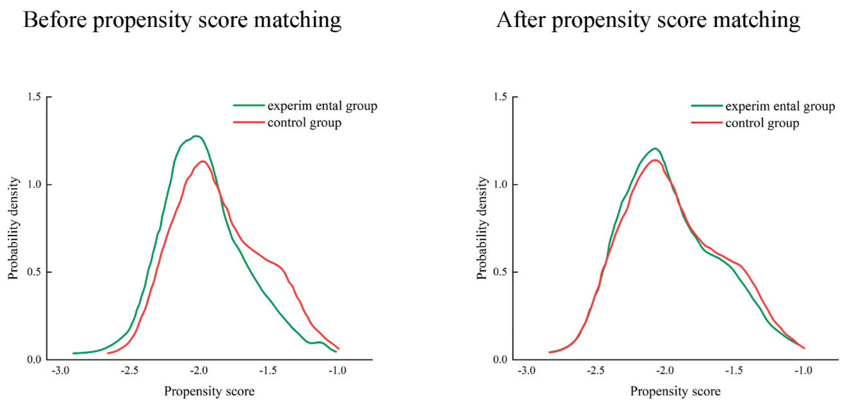

Fig. 2. Kernel density function diagram
Before propensity score matching
After propensity score matching.

Table 5
Balance test results.

Matching variable	Before matching	Mean value		Standard deviation (%)		T-test
	After matching	Experimental group	Control group	Deviation	Reduction	P>ItI
age	U	11.071	10.278	12.2		0.000
	M	11.071	11.082	−0.2	98.6	0.967
Top1	U	38.538	34.903	23.8		0.000
	M	38.538	38.685	−1.0	96.0	0.813
boardshareholdingratio	U	0.09395	0.11708	−12.7		0.000
	M	0.09395	0.09248	0.8	93.6	0.834
ROA	U	0.04087	0.03928	2.9		0.335
	M	0.04087	0.04058	0.5	81.8	0.895
leverage	U	0.46359	0.43345	14.0		0.000
	M	0.46359	0.4617	0.9	93.7	0.826
lnintangibleassets	U	19.087	18.491	33.7		0.000
	M	19.087	19.112	−1.4	95.8	0.724
lnsize	U	22.553	22.1	33.9		0.000
	M	22.553	22.518	1.22.7	92.2	0.526
ToverTlY	U	3.7808	4.1961	−15.1		0.000
	M	3.7808	3.7663	0.5	96.5	0.892

Table 6
Benchmark regression results.

Variable	Stock price fluctuation
treated×time	0.0033194***
	(3.01)
age	−0.0030869
	(-1.20)
Top1	0.0000291
	(0.74)
boardshareholdingratio	0.0080835**
	(2.16)
ROA	−0.0011947
	(-0.21)
leverage	0.0004077
	(0.17)
lnintangibleassets	−0.0003477
	(-0.97)
lnsize	−0.0012297**
	(-1.73)
Idcode	control
Year	control
_cons	0.9675222***
	(37.33)

4.2.5. Further analyses

Since China's transition from a planned economy to a socialist market economy in 1978, the improvement of regional financial development has become an important measure of marketization. Existing research has confirmed that the external macro environment, such as the stage of financial development, has an impact on the political connections between company resource acquisition and utilization. In areas with lower levels of financial development, private enterprises are more motivated to seek political connections as a substitute for formal procedures due to difficulties in financing and accessing other resources. The prospect of private enterprises disclosing "bad news" reduces the risk of stock price fluctuations and even crashes, promoting the stability of stock prices. The lack of political connections has prompted many businesses in areas with higher financial development to stop relying on government development and instead rely on self financing. The level of corporate debt financing has rapidly increased [42], and the potential for venture capital is higher in regions with higher levels of financial development. These factors make it easier for companies that cut off political ties to obtain financing. In addition, the improvement of financial development level helps to regulate the external institutional environment to a certain extent, improving the transparency of enterprise information disclosure. Due to the increase in debt level, the conflict of interests between creditors and shareholders intensifies. In order to further explore the relationship between the loss of political connections and stock price fluctuations, this study examined the moderating effect of financial development level on this relationship.

The model setting structure of the adjustment effect test of the level of financial development [41] is shown in Formula (4):

$$VOL_{it} = \beta_0 + \beta_1 treated_{it} \bullet time_{it} + \beta_2 treated_{it} + \beta_3 time_{it} + \beta_4 control_{it} + \beta_5 treated_{it} \bullet time_{it} \bullet FD_{it} + \beta_6 FD_{it} + \beta_7 Ind_{it} + \beta_8 Year_{it} + \varepsilon_{it} \tag{4}$$

Table 7
Moderating effects.

Variable	Stock price fluctuation
treated×time×MF	−0.0000102**
	(-1.97)
treated×time	0.0054935***
	(3.54)
age	−0.0029272
	(-1.14)
Top1	0.0000342
	(0.86)
boardshareholdingratio	0.0079147**
	(2.12)
ROA	−0.0022684
	(-0.41)
leverage	0.0002499
	(0.10)
lnintangibleassets	−0.0003544
	(-0.99)
lnsize	−0.0013697*
	(-1.92)
MF	5.61e-06***
	(2.67)
ToverT1Y	0.003639***
	(-1.92)
Idcode	control
Year	control
_cons	0.9686732
	(37.38)

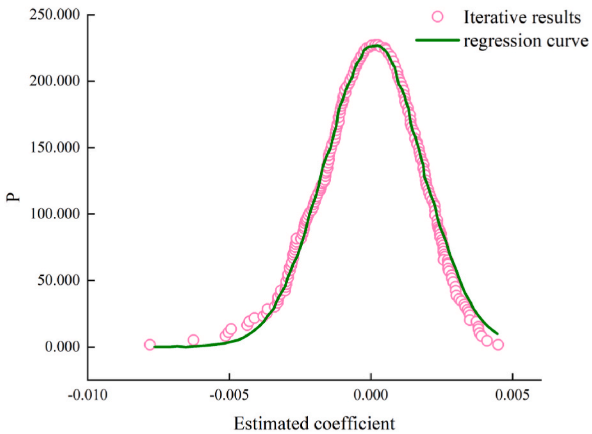

Fig. 3. Placebo testing.

Formula (4) *FD* represents the regional financial development level, which is measured by the regional financial development index. According to Wang [43], the ratio of the balance of deposits and loans held by financial institutions in each province is used in the calculation, and the formula is shown in Formula (5). The moderating impact of media attention on the deterioration of political ties and stock price volatility [42] is represented by the coefficient of interaction term.

$$FD = \frac{\text{Balance of deposits and loans of financial institutions}}{\text{GDP}} \tag{5}$$

This study introduces the interaction term between the loss of political connection and the level of regional financial development, regresses the model, and tests whether the level of regional financial development has a moderating effect on the loss of political connection and the volatility of corporate stock prices (4). Table 8 displays the results of the regression. The results show that the three interaction coefficients are significantly positive at the 5% level ($\beta_5 = 0.0015512$, $p < 0.05$), indicating that in areas with a high level of financial development, the loss of corporate political connections has a higher impact on stock price volatility [44].

Table 8
Moderating effects.

Variable	Stock price fluctuation
treated×time×FD	0.0015512**
	(2.51)
treated×time	−0.021044**
	(-0.84)
age	−0.0029272
	(-1.14)
Top1	0.0000342
	(0.86)
boardshareholdingratio	0.0081497**
	(2.18)
ROA	−0.0007404
	(-0.13)
leverage	0.0002499
	(0.10)
lnintangibleassets	−0.0004635
	(-0.93)
lnsize	−0.001225*
	(-1.72)
FD	0.0015512
	(1.41)
ToverT1Y	0.0004103***
	(3.60)
Idcode	control
Year	control
_cons	0.9604551***
	(36.86)

5. Research conclusions and Implications

5.1. Research conclusions

This study constructed a quasi natural experiment based on the 2013 Central Organization Department's "No.1 Document" and the resignation events of "official" independent directors. On the basis of controlling for sample selection bias and endogeneity, this study used propensity score matching (PSM) method to empirically test the impact of political affiliation loss on stock price volatility, and further explored the role of media attention. Research has found that China's capital market is significantly influenced by government policies, and the deterioration of corporate political connections has a positive impact on stock price volatility. This indicates that political connections have played a role in bringing various benefits to businesses. In addition, independent directors as officials will also take measures to reduce stock price volatility to protect their reputation. These two parties work together to maintain the stability of the company's stock price. The loss of political connections leads to the disappearance of external resources and a favorable institutional environment, which significantly increases the volatility of corporate stock prices.

Secondly, this study confirms that in situations with high media attention, the impact of political affiliation losses on stock price volatility weakens. This may be due to the methods introduced by the Organization Department of the Communist Party of China Central Committee, which often result in more positive media coverage. The relevant media strives to eliminate their own misunderstanding of information. This may also be due to cautious consideration. After obtaining true information, the management of the company is motivated to adjust its development strategy and take action to strengthen its long-term core competitiveness, which helps to stabilize the company's stock price in this situation. In regions with higher levels of financial development, companies that have lost their political connections tend to seek funding from financial institutions after losing their initial financing advantages or channels. Therefore, their debt levels have increased. In addition to transparency in information disclosure it can easily lead to a lack of trust among shareholders, thereby increasing the likelihood of stock price volatility.

5.2. Research implications

The impact of this study on government public policies, corporate stock price stability, and stock market stability is mainly reflected in the following aspects, which have been confirmed by theoretical debates and empirical analysis. Enterprises should establish healthy government enterprise relationships, not overly rely on government resources for financing, but focus on formulating and implementing long-term development goals to enhance their core competitiveness. In the event of an unexpected situation, enterprise strategy makers must correctly analyze various factors including enterprise risks and financing environment, rather than blindly responding to so-called "difficulties" through external financing. The government should guide the media on how to accurately report on the facts of economic, industrial development, and commercial operations, as well as how to position itself and actively participate in economic and social activities. In addition to reducing the risk of a sharp drop in corporate stock prices, the government should also strive to prevent serious harm to investor interests.

There are many other shortcomings in this study. To ensure the effectiveness of the research, independent directors such as representatives of the Chinese family, members of the Chinese People's Political Consultative Conference, and government agencies are not considered "official" independent directors. However, this does not completely rule out the fact that National People's Congress deputies, members of the Chinese People's Political Consultative Conference, and staff of state institutions have political influence on enterprises. In other words, independent directors with such identities may to some extent provide political resources for enterprises. Therefore, after the release of the "18th National Congress" document by the Central Organization Department, relevant enterprises also felt the loss of political ties. Therefore, the sample may not be comprehensive. In addition, this study only focuses on the initial response of the capital market to the loss of political connections. However, this study did not explore the response measures of various stakeholders after the loss of political ties, as well as the specific impact mechanisms of resource loss and institutional environmental advantages on the volatility of corporate stock prices. In addition, the study did not address the improvement measures taken by companies to compensate for potential losses caused by stock price fluctuations.

Availability of data and material

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Qiong Sun: Writing – original draft, Resources. **Qingzhou Xiao:** Writing – original draft, Investigation. **Jingjing Jiang:** Writing – review & editing, Funding acquisition, Data curation. **Xiankai Huang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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